MANCHESTER 1824
The University of Manchester

# Understanding the Earth

## Week 2 Methods of Stratigraphic Correlation using Relative Age Dating

The aims of this week's workshop are:

- to explore how geologists reconstruct geologic history using lithostratigraphic and biostratigraphic methods
- to compare the subdivision of geologic time using relative and absolute dating methods
- to introduce the geologic time scale and its major subdivisions

# Introduction to lithostratigraphy

*Recap from last week's workshop:*

Law of superposition: in any succestion of strata (rock layers) that has not been overturned since being deposited, younger rocks lie on top of older rocks.

Contacts: planar or irregular surfaces that separate diffeternt rock units (lithologies). Contacts can be abrupt or gradational.

Unconformity: a surface of erosion or non-deposition separating youger strata from older rocks that represents a signbificant gap in time.

Lithostratigraphy subdivides the rock record into bodies of rock that have specific lithological characteristics and stratigraphic position relative to other bodies of rock.

# Hierarchy of lithostratigraphic units

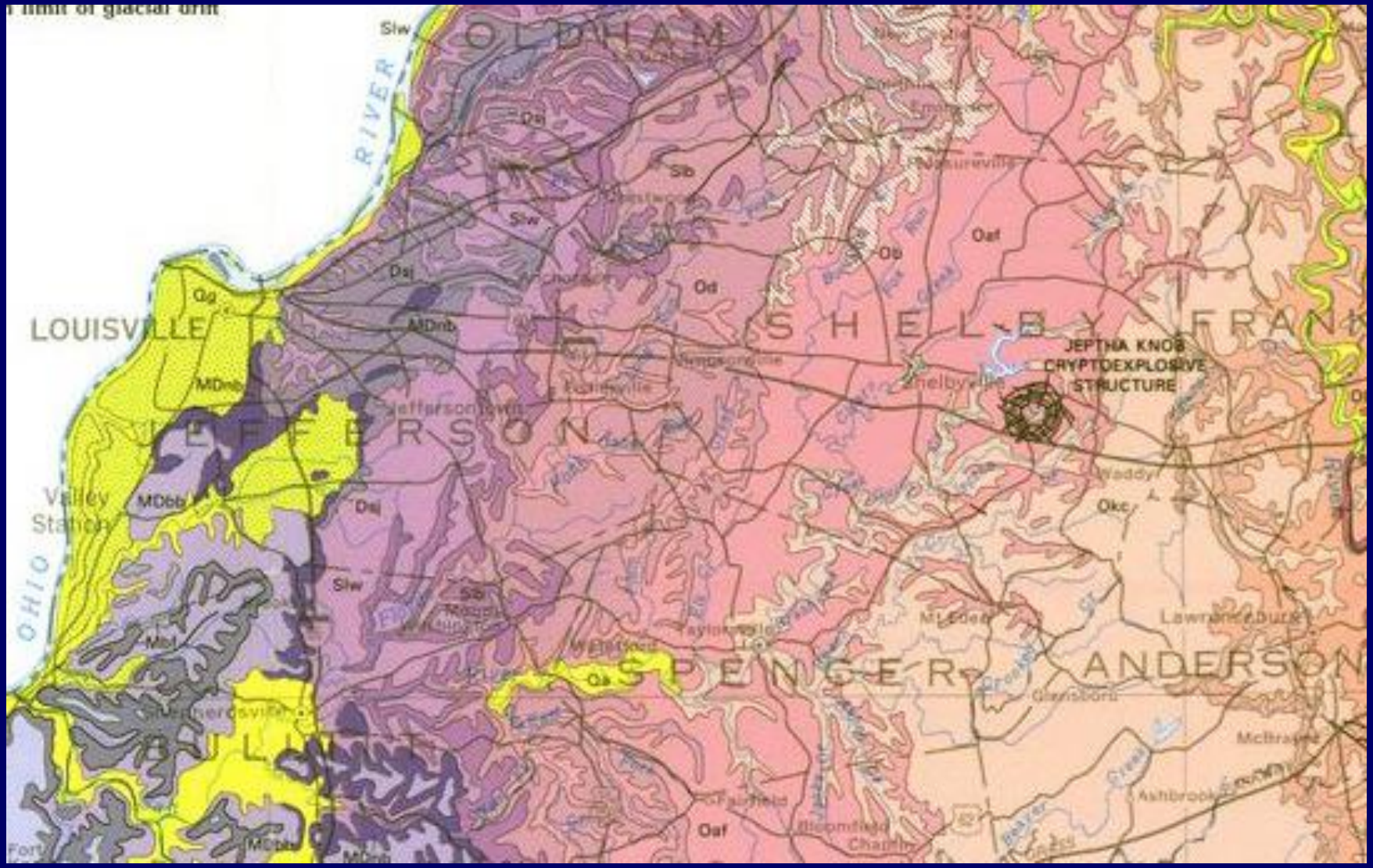



| | |
|---|---|
| Supergroup | A collection of related groups that is bound by regionally persistent unconformities |
| Group | A grouping of two or more formations that share certain characteristics, commonly bounded by laterally persistent unconformities |
| **Formation** | **The basic unit - a body of material that can be identified by its lithological characteristics and by its stratigraphic position; must be laterally persistent (mappable), and have clearly defined lithological characteristics (*e.g.*, mineralogy, texture sedimentary structures, fossil content)** |
| Member | A unit with limited lateral extent but which is consistently related to a particular formation (*e.g.*, a conglomeratic interval within a formation that is dominantly sandstones) |
| Bed | A distinctive unit within a formation or member |

# Exercise 1: Introduction to lithostratigraphy



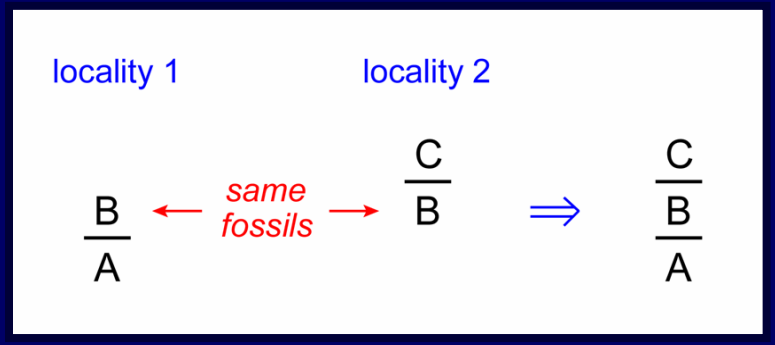
# Introduction to lithostratigraphy

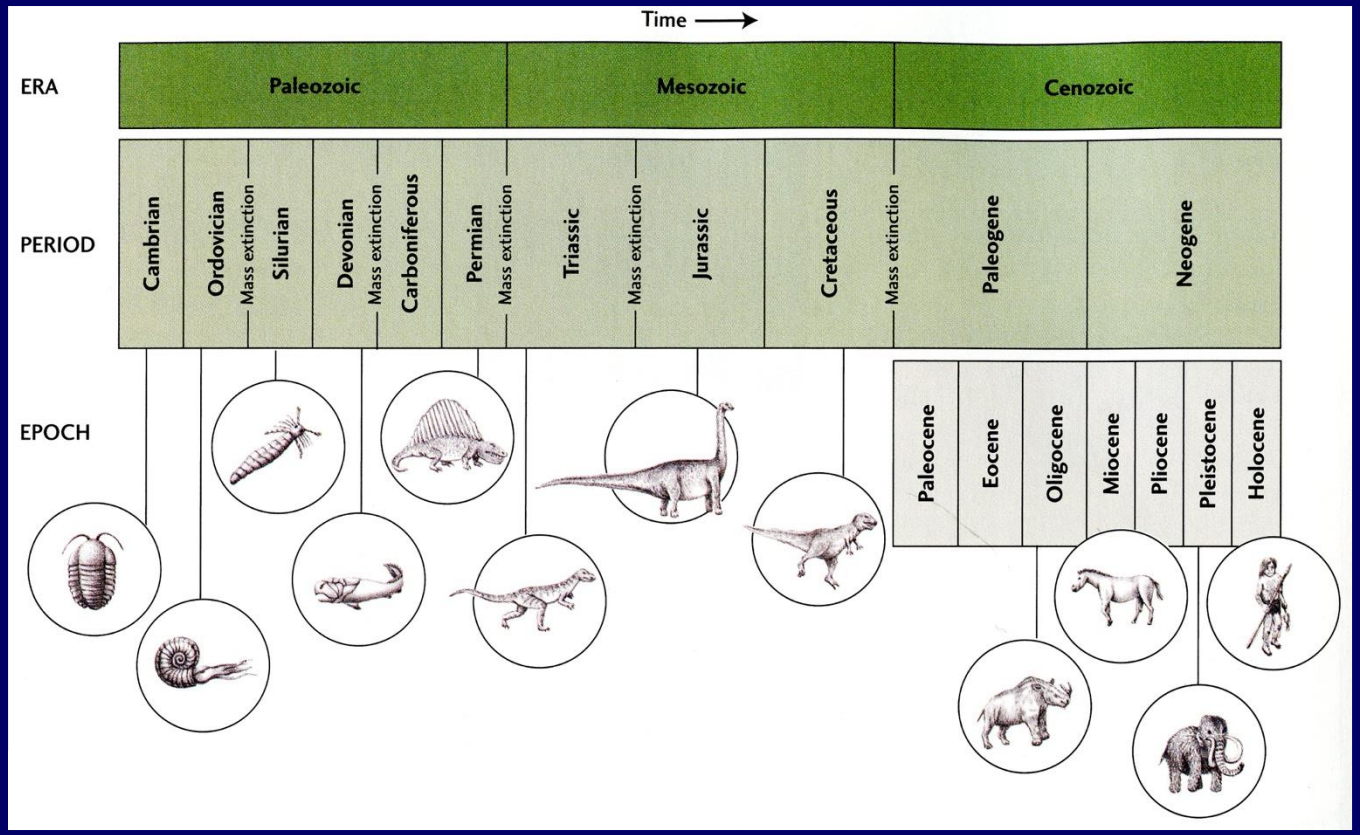


# The use of fossils in stratigraphy

To build up a picture of Earth history we need to match up events between localities

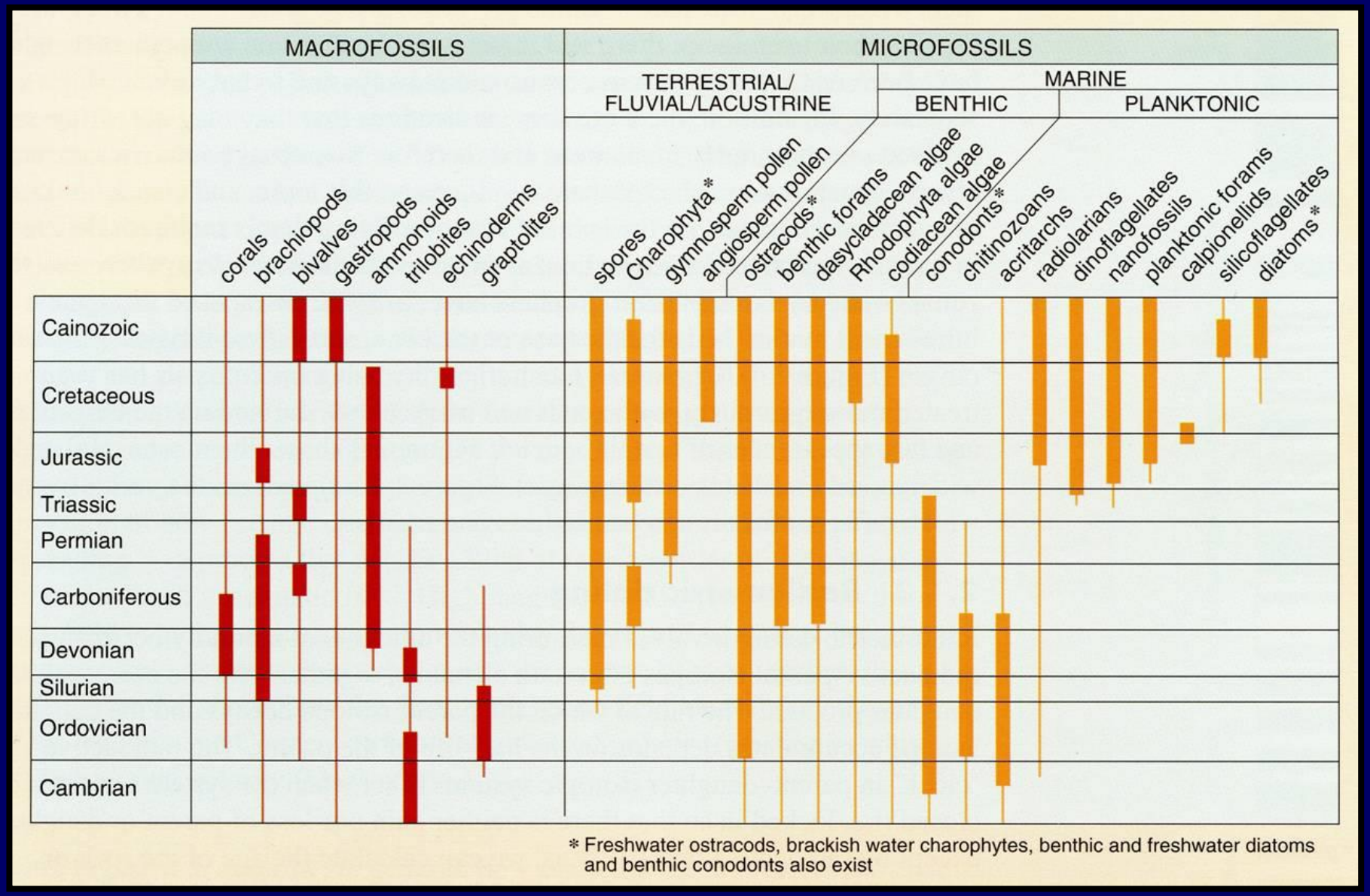
Some unifying principle is required. In part, this comes from the further study of fossils




N=50, N=82 and N=126

# Biostratigraphy



# Exercise 2: Faunal succession




Road cut where the Interstate 64 crosses Clear Creek, east of Louisville
Northern Kentucky, USA

# Fossils collected



A. The crinoid
*Ectenocrinus*
(Fulton-Fairmount)



B. The trilobite
*Isotelus*
(Fulton-Amheim)



C. The brachiopod
*Zygospria*
(McMicken-Fairmount)



D. The gastropod
*Cyclonema*
(McMicken-Fairmount)





E. The brachiopod
*Vinlandostrophia*
(Fairmount-Corryville)

I. The clam
*Ambonychia*
(Fairmount-
Bellevue)





F. The bryozoan
*Parvohallopora*
(Southgate-Fairmount)

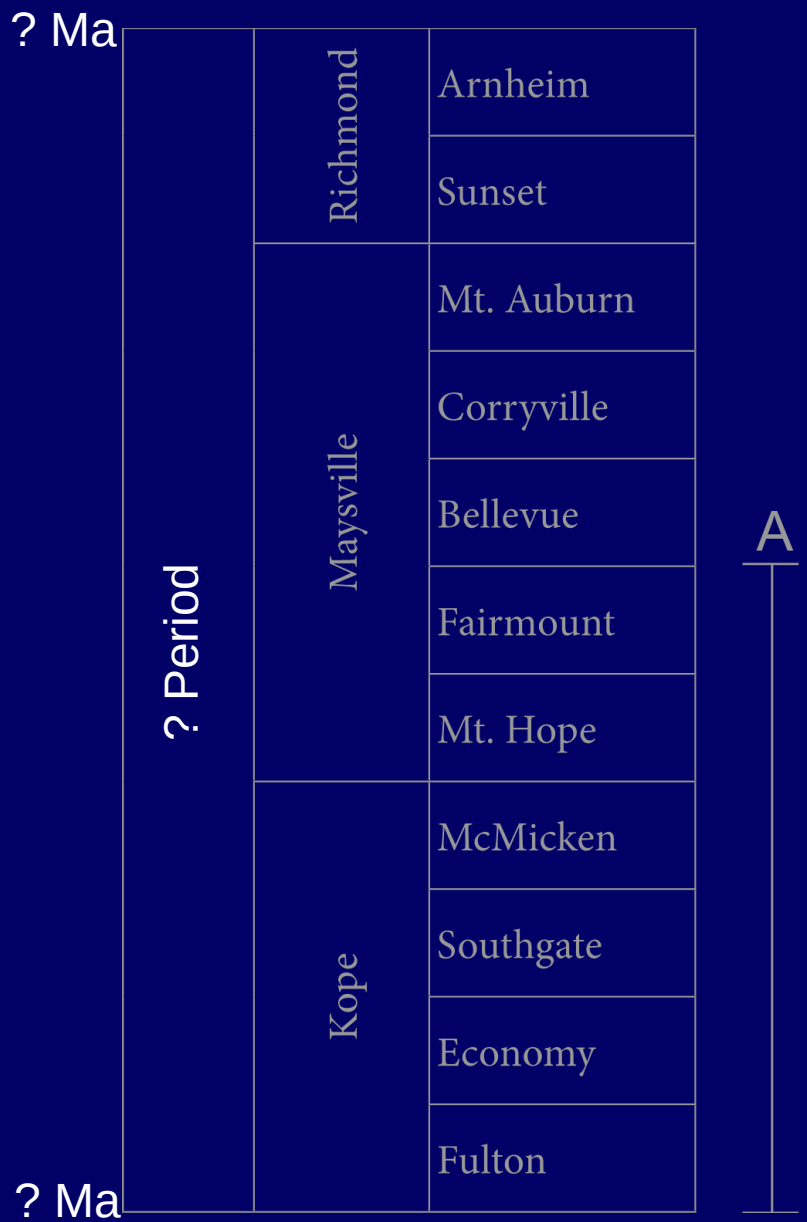



G. The brachiopod
*Cincinnetina* (Fulton-Fairmount)



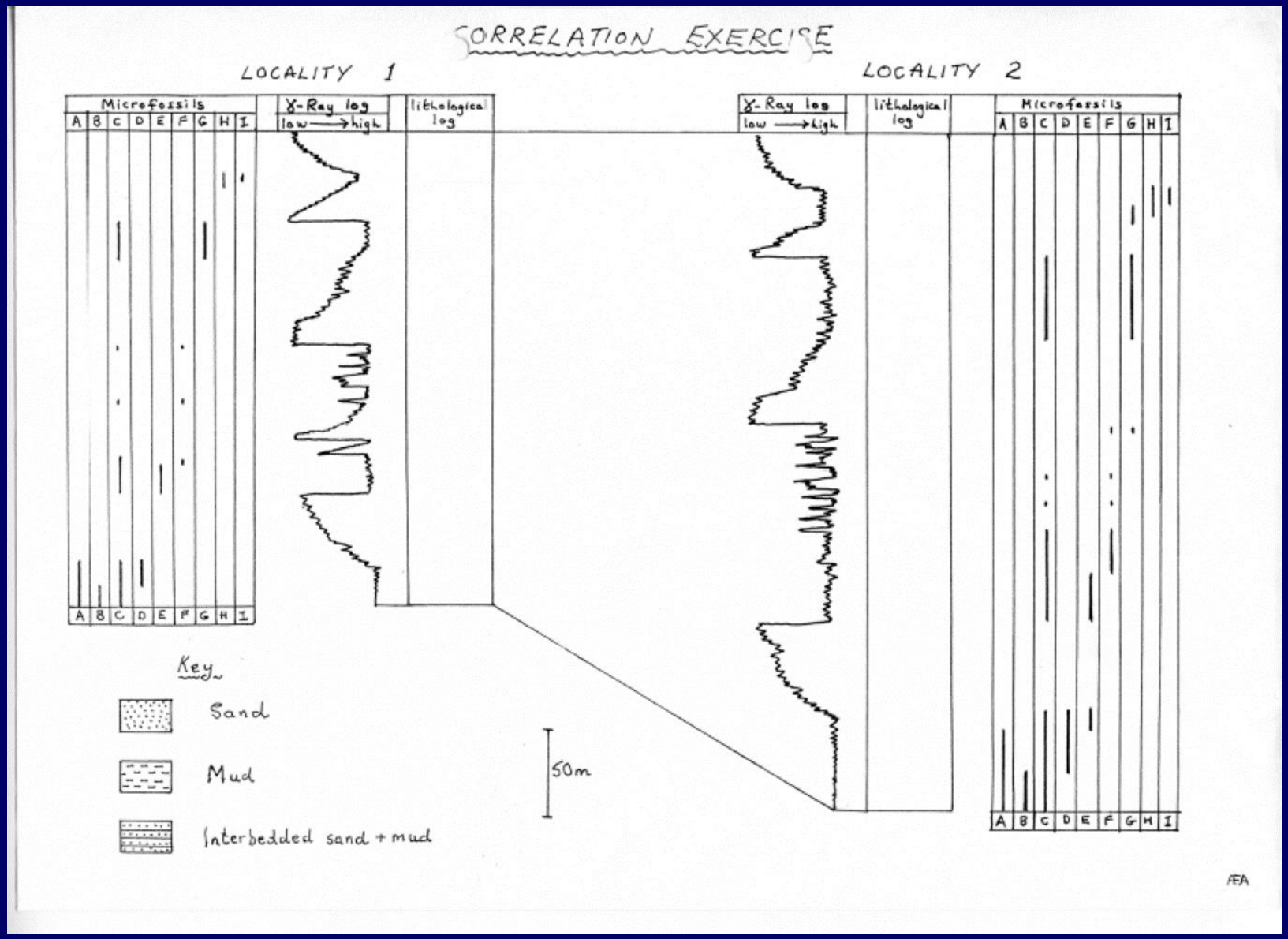
H. The edrioasteroid
*Streptaster*
(Southgate-Sunset)

# Practice 2 Geologic range of fossils collected




Plot the geologic range of the remaining eight animals to narrow down when the rock was formed and prepare answers questions 1-3

# Exercise 3: Correlated lithostratigraphy



# The division of geological time today





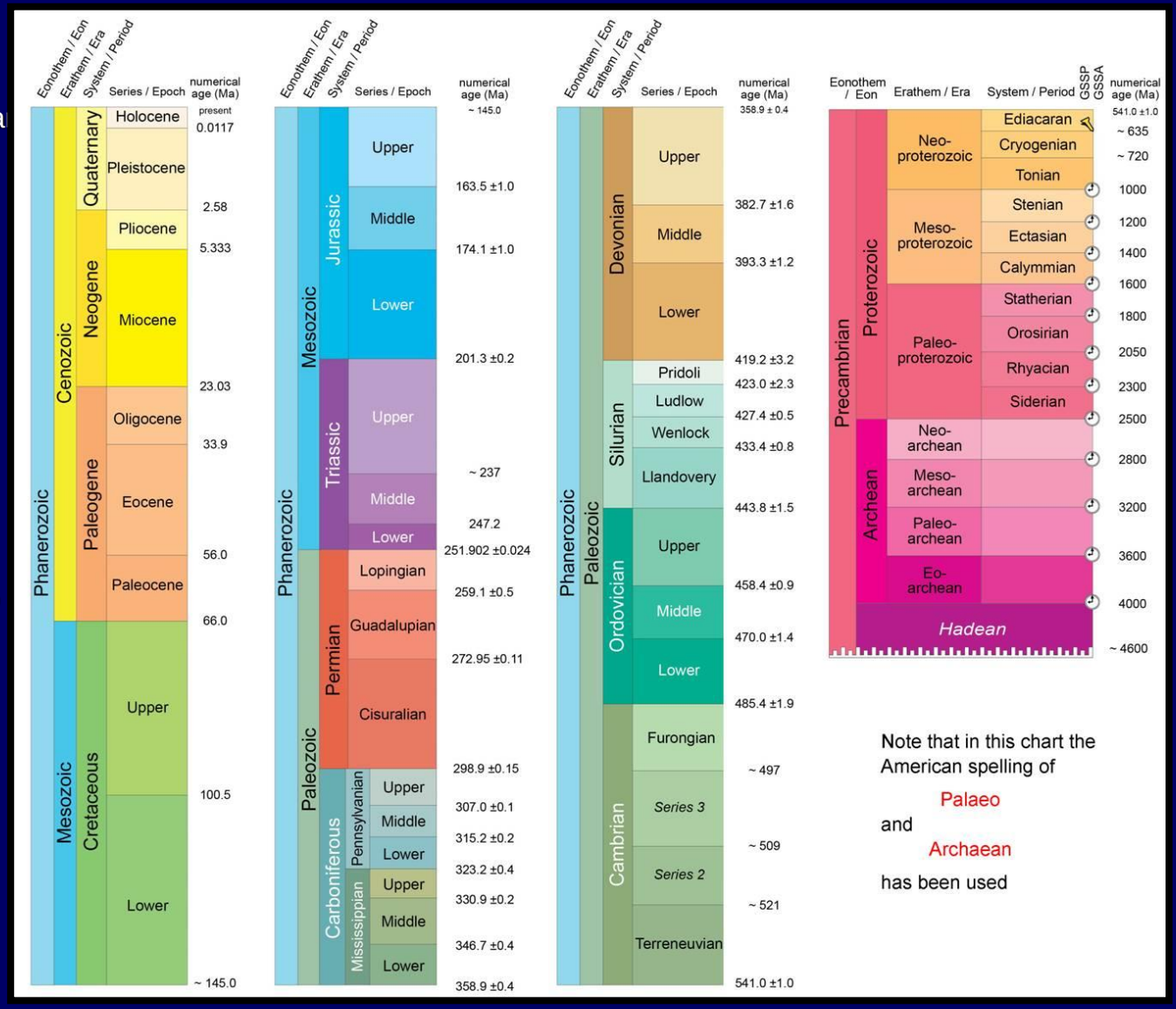

The golden spike defining the base of the Ediacaran Period (635 Ma), Enorama Creek, Flinders Ranges, South Australia

Global Stratotype Section and Point (GSSP) *aka a* "golden spike"

- define the base of geochronological units
- physical point within a succession of strata at a particular place somewhere in the world
- rocks at GSSP dated (i.e. located in an absolute time scale)

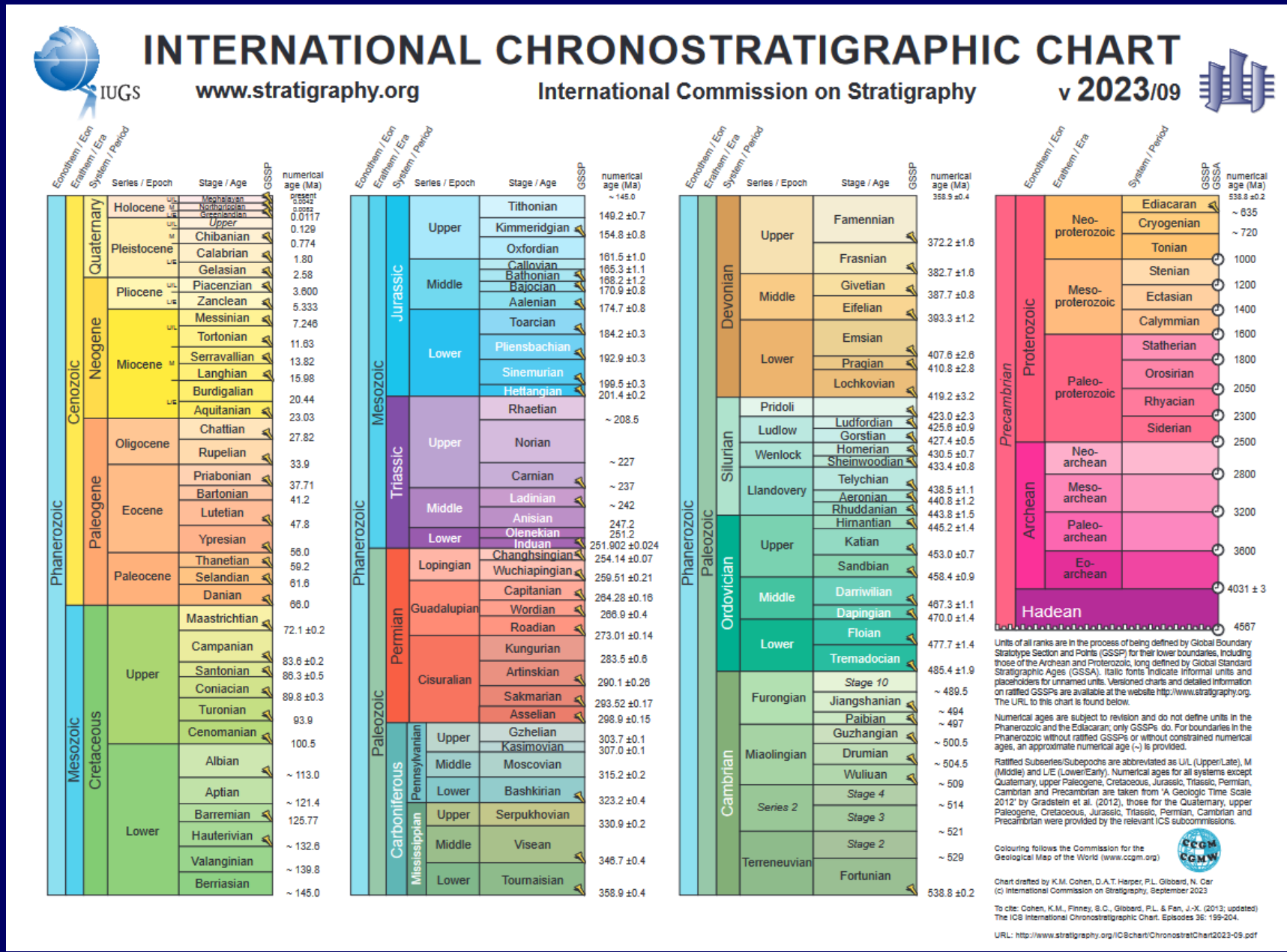
# Subdivisions of geochronological units

1. **Eons**
   Archaean, Proterozoic, Phanerozoic
   ↕ Precambria
2. **Eras**
   (Precambrian eras), Palaeozoic, Mesozoic Cenozoic
3. **Periods**
   Cambrian, Ordovician, *etc.*
4. **Epochs**
   Some named, others Lower / Middle / Upper
5. **Ages**
   Smallest commonly used unit




www.stratigraphy.org

# The 2023 geological timescale (look up the latest version)



# Can you do what we set out to do?

On successful completion of the workshop you will be able to:

- Identify rock units with different lithological characteristics
- use fossils to date a rock unit by applying principles of faunal succession
- distinguish between relative and absolute dating methods
- summarise how geologists created and refined the geologic time scale